

Scalability & Future Plan

Date	3 July 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Credit Card Approval Prediction project can be enhanced to support larger datasets, increased user traffic, improved prediction performance, and advanced deployment options. It also presents a roadmap for extending the solution with additional intelligent features and cloud-based services.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application is designed for a single-user environment and cannot efficiently support multiple concurrent users.	Reduced performance under heavy usage.	High
2	The model is trained on a single dataset and may require retraining to maintain accuracy with new or changing data.	Prediction accuracy may decrease over time.	High
3	The application is currently deployed locally and is not accessible over the internet.	Limited accessibility and scalability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single local user.	Deploy the application on a cloud platform to support multiple simultaneous users.
2	Data Storage	Uses a static CSV dataset.	Integrate a relational database such as MySQL or PostgreSQL for efficient data management.
3	Performance	Predictions are generated using a locally stored trained model.	Optimize the model, implement caching, and improve response time for larger workloads.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS encryption, secure API access, and input sanitization for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform for public access.	Short Term	Improved accessibility and availability.
Phase 2	Integrate a database for secure storage and management of applicant information.	Medium Term	Better scalability and efficient data handling.
Phase 3	Implement advanced machine learning techniques and periodic model retraining to improve prediction accuracy.	Medium Term	Increased prediction performance and reliability.
Phase 4	Develop REST APIs and a mobile-friendly interface to support multiple platforms and third-party integration.	Long Term	Enhanced usability, flexibility, and broader adoption.